



Experiment 2: DC Sweep Analysis

Electronics Engineering Department
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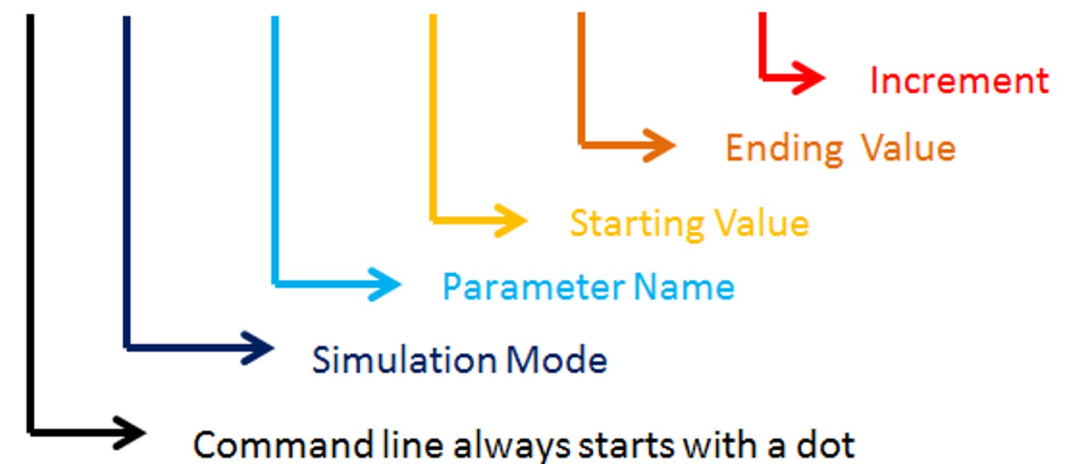


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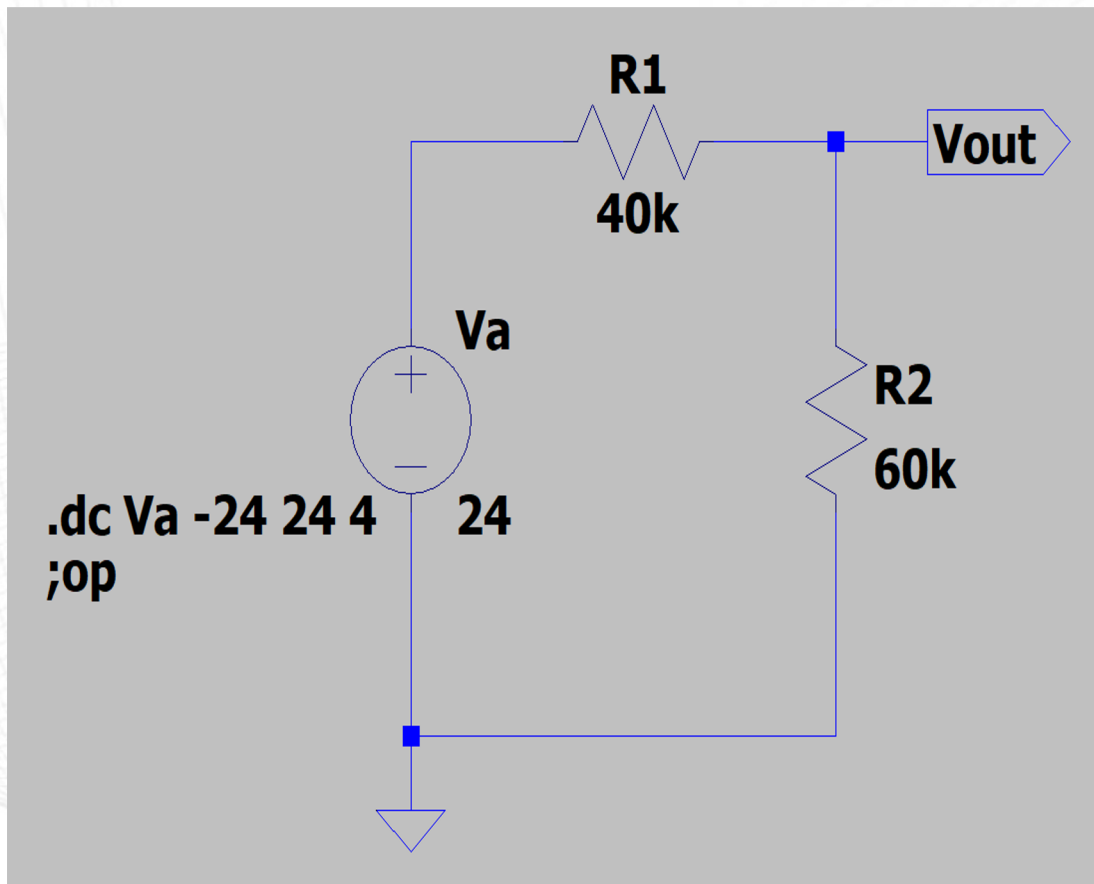
Simulation: DC Sweep

- Allows the user to vary the voltage or current of a specific device.
- It can be single or nested

.dc V1 0 10 0.1

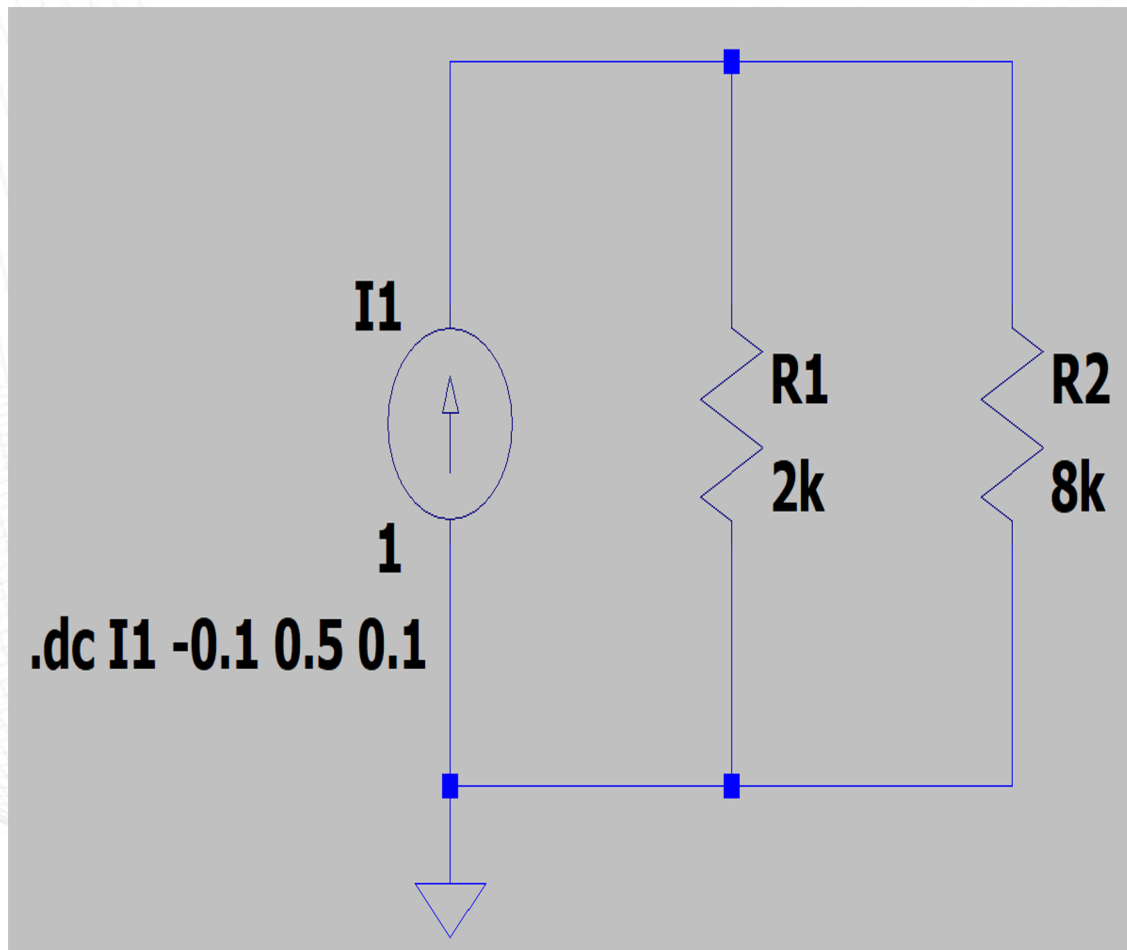


Circuit #1



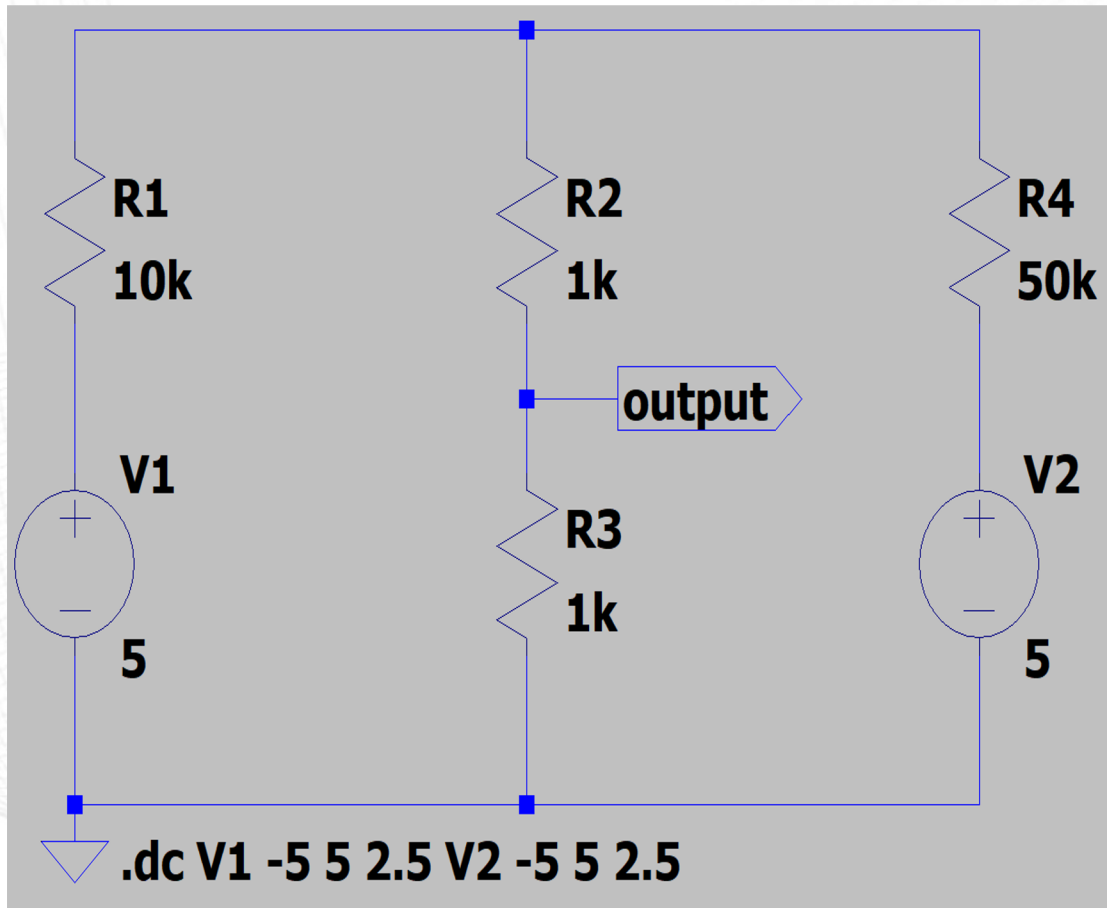
- Examine the output voltage and current in the circuit for different values of V_a .
- Examine the power dissipated in R_2 for different values of V_a .
- What is the resulting expression generated by LTSPICE to compute for the power?
- Think of other ways to generate the same graph without using ALT + left-click.

Circuit #2



- Examine the current in R_1 for different values of I_1 .
- Examine the current in R_2 for different values of I_1 .
- Examine the voltage across R_2 for different values of I_1 .
- Examine the power dissipated in R_1 .
- Examine the power dissipated in R_2 .

Circuit #3



- Examine the current in R3.
- Examine the output voltage.
- Examine the power dissipated in R3.
- What are the combination values of V1 and V2 that will result to a +100mV output voltage?

References:

- **Experiment Paper in ECE2113 by:**
Engr. Mary Grace Ann Bautista, M.Sc.
Engr. Mark O. Montances, M.Sc.
- **Photo courtesy of: ElectronicsBeliever**